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Evaluation of Identity and Access Management (IAM) Weaknesses in Cloud Platforms

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Abstract

Cloud computing has transformed the way organizations store data, deploy applications, and provide online services. As businesses increasingly migrate to cloud platforms, protecting digital identities and controlling access to cloud resources have become essential. Identity and Access Management (IAM) is a security framework that ensures only authorized users can access cloud resources based on predefined permissions. Although cloud providers offer advanced IAM services, organizations often misconfigure these services, resulting in severe security vulnerabilities.

This case study evaluates the common IAM weaknesses found in cloud platforms and demonstrates how poor identity management can lead to unauthorized access, privilege escalation, data breaches, and financial loss. It also discusses best practices and security recommendations to strengthen IAM implementations.

1. Introduction

Cloud computing provides organizations with flexible, scalable, and cost-effective computing resources. Popular cloud providers such as Amazon Web Services (AWS), Microsoft Azure, and Google Cloud Platform (GCP) offer Identity and Access Management systems that help administrators control who can access cloud resources.

IAM is responsible for authentication, authorization, user management, role assignment, and permission control. When IAM policies are correctly configured, organizations can effectively protect sensitive data and critical infrastructure. However, poor configuration, excessive privileges, and weak authentication mechanisms frequently become the primary cause of cloud security incidents.

This case study investigates IAM weaknesses through a real-world scenario and evaluates their impact on cloud security.

2. Overview of Identity and Access Management (IAM)

Identity and Access Management is a framework of policies, technologies, and procedures used to manage digital identities and control access to computing resources.

The main objectives of IAM include:

- Authenticating user identities.
- Authorizing access based on roles.
- Enforcing least-privilege access.
- Monitoring user activities.
- Protecting sensitive cloud resources.

Core IAM Components

Component	Description
Authentication	Verifies user identity
Authorization	Determines user permissions
Roles	Group of permissions assigned to users
Policies	Rules controlling access
Multi-FactorAuthentication (MFA)	Additional verification layer
Identity Federation	Allows login using external identity providers

Proper implementation of these components significantly reduces the risk of unauthorized access.

3. Case Study Scenario

Organization

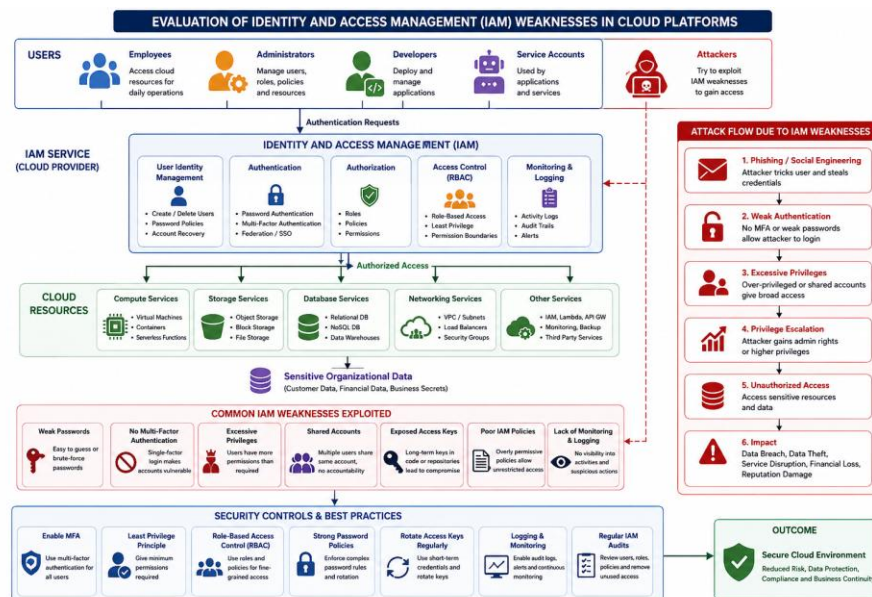
ABC Financial Services is a medium-sized financial company that migrated its customer database and internal applications to a cloud platform.

The organization created IAM users for employees and administrators. Due to rapid migration and poor security planning, several IAM weaknesses remained unnoticed.

Existing Configuration

- Administrator account shared among multiple employees.
- No Multi-Factor Authentication enabled.
- Several users assigned Administrator privileges.
- Long-term access keys stored in application code.
- Password policy disabled.
- IAM roles not regularly reviewed.
- No activity logging for privileged accounts.

Initially, the system operated without noticeable issues. However, these weaknesses eventually resulted in a serious cloud security incident.



4. IAM Weaknesses Identified

4.1 Excessive User Privileges

Many employees received AdministratorAccess permissions even though they only required read-only access.

This violated the Principle of Least Privilege and significantly increased security risks.

Possible consequences include:

- Data deletion
- Service disruption
- Resource modification
- Unauthorized account creation

4.2 Absence of Multi-Factor Authentication

- The administrator account relied only on username and password authentication.
- An attacker successfully obtained administrator credentials through phishing.
- Without MFA, the attacker gained complete control over the cloud environment.

4.3 Weak Password Policy

The organization allowed weak passwords such as:

- Password123
- Company2024
- Admin123

Weak passwords are vulnerable to brute-force and dictionary attacks.

4.4 Long-Term Access Keys

Developers embedded permanent access keys inside application source code.

After the code was uploaded to a public repository, attackers discovered the credentials and accessed cloud resources.

4.5 Shared Administrator Account

Multiple employees shared a single administrator account.

This created accountability issues because security teams could not determine who performed specific administrative actions.

4.6 Lack of Logging and Monitoring

Cloud activity logs were not enabled.

The attacker remained undetected for several weeks while downloading sensitive customer information.

5. Attack Scenario

An attacker first launched a phishing campaign targeting cloud administrators.

One employee unknowingly entered cloud credentials into a fake login page.

Since Multi-Factor Authentication was not enabled, the attacker immediately logged into the cloud environment.

Because the compromised account possessed Administrator privileges, the attacker was able to:

- Create new administrator accounts.
- Disable security alerts.
- Download confidential customer databases.
- Delete activity logs.
- Launch cryptocurrency mining instances.

The organization discovered unusual cloud billing charges several weeks later.

By that time, sensitive financial information had already been compromised.

6. Risk Assessment

Weakness	Risk Level	Business Impact
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No MFA	High	Unauthorized access
Excessive Privileges	High	Complete cloud compromise
Weak Passwords	Medium	Credential theft
Shared Accounts	Medium	Lack of accountability
Long-Term Access Keys	High	Credential leakage
No Logging	High	Delayed incident detection

7. Security Recommendations

To strengthen IAM security, the organization should implement the following controls.

Enable Multi-Factor Authentication

All privileged accounts should require MFA.

This significantly reduces the likelihood of unauthorized access.

Implement Least Privilege

Users should receive only the permissions required for their job responsibilities.

Administrative permissions should be granted temporarily when necessary.

Use Strong Password Policies

Organizations should enforce:

- Minimum 12-character passwords
- Complexity requirements
- Password expiration
- Password history

Rotate Access Keys

Long-term credentials should be replaced with temporary credentials.

Access keys should be rotated regularly.

Enable Cloud Logging

CloudTrail (AWS), Azure Monitor, or Google Cloud Logging should continuously monitor user activities.

Security teams should review logs regularly.

Conduct Regular IAM Audits

Organizations should periodically review:

- User accounts
- Roles
- Permissions
- Policies
- Unused credentials

Unused accounts should be removed immediately.

8. Best Practices

Organizations should adopt the following IAM best practices:

- Apply the Principle of Least Privilege.
- Enable Multi-Factor Authentication.
- Use Role-Based Access Control (RBAC).
- Rotate credentials regularly.
- Remove inactive users.
- Monitor login activities.
- Perform periodic security audits.
- Encrypt sensitive data.
- Enable continuous monitoring.
- Conduct employee security awareness training.

Following these practices significantly improves cloud security.

9. Conclusion

Identity and Access Management is one of the most important security components in cloud computing. Even though cloud providers offer advanced IAM services, organizations frequently introduce vulnerabilities through poor configuration and weak security practices. The case study demonstrated how excessive privileges, weak passwords, shared administrator accounts, disabled Multi-Factor

Authentication, and inadequate monitoring can allow attackers to compromise cloud environments.

Organizations should adopt a Zero Trust approach, enforce least-privilege access, implement strong authentication, and continuously monitor user activities. Regular IAM audits and employee awareness programs further reduce security risks. Proper IAM implementation not only protects sensitive information but also ensures regulatory compliance and strengthens overall cloud security.

References

1. Amazon Web Services. AWS Identity and Access Management (IAM) User Guide.
2. Microsoft. Azure Identity and Access Management Documentation.
3. Google Cloud. Cloud Identity and Access Management Documentation.
4. National Institute of Standards and Technology (NIST). Digital Identity Guidelines (SP 800-63).
5. Cloud Security Alliance. Security Guidance for Critical Areas of Cloud Computing.
6. OWASP Foundation. Identity and Access Management Cheat Sheet.